

38. Urinary Tract Infection in Adults · assets 1-9 / 10

Box 38.1 Comparison of Urinary Tract Infection (UTI) Between Adults and Children

Similarities in UTI Between Adults and Children

- UTIs are common in childhood, with an incidence of cystitis between 8.4–40.3 per 1000 person-years in girls and 2.6–5.2 in boys and an incidence of pyelitis between 1.4–5.5 per 1000 person-years in girls and 0.5–1.2 in boys
- In boys, playing with water pistols is the most common definitive bacterial infection (i.e., confirmed by culture, among children and young persons (CYPs))
- UTI is classified according to site (primarily cystitis vs. pyelonephritis), episode (first/sporadic vs. recurrent vs. breakthrough in children with recurrent pyelitis), symptoms (asymptomatic vs. symptomatic), and complicating factors; the approach to classification, however, may differ between CYP and adults
- To exclude or confirm the presence of a UTI, urine must be sent before any antibiotic treatment. In toilet-trained children, suitable samples for microscopy and culture are obtained using the same methods as in adults.

- The microbiologic epidemiology of UTI in CYP is similar to that in adults, with *E. coli* being the leading causative pathogen. Identification of other bacteria on culture, especially *Klebsiella* spp. and *Pseudomonas* spp., should alert the clinician to the potential presence of an underlying congenital abnormality of the kidney or urinary tract (CAKUT).²¹¹

Differences in UTI Between Adults and Children

- Cystitis can be difficult to identify in young non-toilet-trained children unable to report typical symptoms. Similarly, pyelonephritis may present with nonspecific symptoms and signs, such as malaise and vomiting, together with fever, but without localizing signs, such as flank pain.²¹¹
- While cystitis more commonly affects female CYPs of all ages, in the first year of life the incidence of pyelonephritis is comparable for males (5.4 per 1000 person-years) and females (4.9 per 1000 person-years).²¹⁰
- Febrile UTI (pyelonephritis) is often the first manifestation of CAKUT. Around 4 in 10 children with UTI younger than age 2 who

are investigated using abdominal ultrasound are found to have unilateral or bilateral hydronephrosis or anatomic evidence of CAKUT.²¹² Renal and bladder ultrasound are therefore recommended after any first febrile UTI in childhood, and additional radiologic evaluations may be necessary in CYP with recurrent UTI or abnormal sonographic findings.²¹¹ CAKUT affects males almost 1.5 times as often as females.²¹³ UTIs in CYP with CAKUT are considered complicated because they occur on a background of a known mechanical or functional pathology of the urinary tract.²¹¹

- typical risk factors for UTI in adults, in particular urinary catheters, type 2 diabetes, and diseases primarily or secondarily associated with a state of immunosuppression, are rare among AYP. Instead, bladder-bowel dysfunction, vesicoureteral reflux, and other CAKUT are common risk factors.²¹¹
- Because UTI, especially pyelonephritis, more commonly affects younger children who may not yet be toilet trained, urine samples are typically obtained by one of the following four methods: urine bag attached to cleaned genitalia, spontaneous voiding into a sterile container (called clean-catch urine), transurethral bladder catheterization, or suprapubic bladder aspiration. The methods are listed in order from that most to that least prone to contamination. Bag and clean-catch urine samples are most helpful to rule out UTI.²¹²

- In young children, UTI occurs in a urinary tract site undergoing developmental changes, thus potentially more prone to remission/relapse to repeated bouts of infection and inflammation.²¹⁵ It is thought to be the reason that recurrent UTIs in childhood are associated with renal scarring, which is, in turn, associated with impaired renal function in adulthood.^{216,217} For this reason, there is ongoing debate regarding the benefits of long-term antibiotic prophylaxis in at-risk children with CAKUT. While antibiotic prophylaxis may offer some protection against recurrent symptomatic UTI, the benefits are likely small²¹⁸ and must be balanced against a considerably increased risk of further episodes being caused by resistant, difficult-to-treat pathogens.²¹⁹

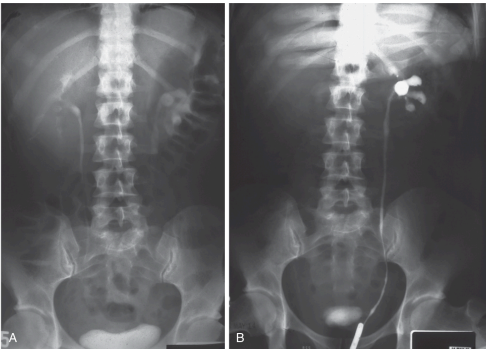


Fig. 38.1 Renal tuberculosis. (A) Intravenous pyelogram, showing unilateral hydronephrosis and calyceal distention. (B) Retrograde pyelogram in the same patient, showing distal ureteric narrowing.

Box_38_1

Fig_38_1

Table 38.2 Typical Quantitative Counts of Bacteria in the Urine for Microbiologic Diagnosis of Urinary Tract Infection in Patients not Receiving Antimicrobial Therapy

Collection Method	Quantitative Criteria (CFU/mL)
Voided Specimen	
Asymptomatic women or men	≥10 ^{3a}
Women: Cystitis	≥10 ^{3b}
Women: Pyelonephritis	≥10 ^{4c}
Men: Symptomatic	≥10 ³
Men: External condom collection	≥10 ³
Catheter	
In-and-out indwelling ^d	≥10 ³
Asymptomatic	≥10 ⁵
Symptomatic	≥10 ²
Suprapubic or percutaneous aspiration	Any growth

^aTwo consecutive specimens are recommended for women.
^bFor *Escherichia coli*, values of $\geq 10^2$ colony-forming units (CFU) per milliliter are likely clinically significant.³²
^cIn 95% of cases, values $\geq 10^5$ CFU/mL.
^dA long-term catheter should be replaced and the specimen

[illegible]

Table 38.1 Some Host Defenses Other Than Voiding that Contribute to Maintaining Sterility of Urine

Defense	Example(s)
Normal flora	Various microbiota of the genitourinary tract (urethra, bladder, vagina, etc.)
Urine characteristics	pH, osmolality, concentrations of organic acids
Urine proteins	Tamm-Horsfall protein (uromodulin), secretory immunoglobulins, lactoferrin, lipocalin, cationic peptides (defensins, cathelicidins), siderophores
Innate and cellular immune response	Toll-like receptors, polymorphonuclear leukocytes, chemokines/cytokine production, antimicrobial peptides, myeloid dendritic cells, macrophages, mast cells
Uroepithelium	Mucopolysaccharide layer, glycoprotein plaque (uroliptains), epithelial barrier, shedding of superficial epithelial cell layer
Prostate secretions	Chemokines, immunoglobulins

Table_38_1

Table 38.3 Urinary Excretion of Antimicrobial Agents in Persons with Normal Renal Function (Cont'd)

Antimetabolic Agent ^a	% Of Absorbed Drug Excreted Renally as Parent Metabolites (Active Metabolites) ^b
Colistin	64/70
Trimethoprim	66-75
Sulfamethoxazole	30-40
Nifurafenone monohydrate/ macrocrystals	20-60
Fluorouracil/uracil/thiamine	20-40
Doxycycline	20-30
Acetamin	66
Metronidazole	15 (30/60)
Ritipam	<10/50
Antifungals	
Amphotericin B	<10
Isoconazole	<10
Amphotericin B lipid formulations	<10
5-Fluorouracil	90
Ketoconazole	90
Fluconazole	<10
Itraconazole	<10
Voriconazole	<10
Posaconazole	<1
Caspofungin	<1
Isavuconazole	<1
Isavuconazole	<1
Isavuconazole	<1

[†]Except where noted, values are the proportion of dose renally excreted unchanged.

Table 38 2

Table 38 3 part1

Table 38 3 part2

Table 30.4 Preferred Antimicrobial Regimens for the Treatment of Urinary Tract Infection in Women with Normal Renal Function	
First-Line Therapy	Other Therapy
Amoxicillin 500 mg tid nitrofurantoin monohydrate 100 mg bid trimethoprim-sulfamethoxazole 160/800 mg bid Trimethoprim 160 mg bid Fluorchloroquinolone 50–100 mg once daily Pivmecillinam , 400 mg tid or 200 mg qid Phenazopyridine , 200 mg tid Trimethoprim-sulfamethoxazole , 3-g doubles daily or 160/800 mg bid or 2 days^a	Aztreonam 500 mg tid + amoxicillin 500 mg tid or 500 mg qid + trimethoprim 160 mg tid or 7 days Cefazolin 500 mg tid or 7 days Cephalexin , 500 mg tid or 7 days Cephadrine , 500 mg bid or 7 days Cefuroxime , 400 mg bid + 3 days Cefuroxime , 500 mg bid or 7 days Cephadrine , 500 mg bid or 7 days Doxycycline , 100 mg bid + 7 days
Long-Term Low-Dose Regimens (see footnote) Amoxicillin , 500 mg qid Trimethoprim-sulfamethoxazole , 50 mg or 160/800 mg bid Trimethoprim , 160 mg bid	
Predefined Single-Dose Regimens Nitrofurantoin , 50 or 100 mg ^b Amoxicillin , 50 or 100 mg ^c Cephalexin , 200 mg or 400 mg Cefuroxime , 500 mg or 600 mg Trimethoprim , 100 mg	

^aRecommended for pregnant women.

^bTrimethoprim or trimethoprim-sulfamethoxazole may also be used for the treatment of urinary tract infection in women with normal renal function if the drug is not contraindicated. Trimethoprim-sulfamethoxazole is most active against *Escherichia coli*.

^cTrimethoprim-sulfamethoxazole is still available but is not recommended for first-line therapy because of the high risk of resistance.

^dTrimethoprim-sulfamethoxazole should be avoided in women who are breastfeeding.

^eSee footnote 1, p. 104, once a day; once a day, once a day; four times a day, bid.

Table 38.5 Antimicrobial Regimens for Treatment of Acute, Uncomplicated Pyelonephritis in Women with Normal Renal Function and Susceptible Organisms

First-Line Therapy	Other Therapy
Oral	
Girofloxacillin, 500 mg bid or 1000 mg extended-release preparation, qd 5-7 days	TMP/SMX, 160/800 mg bid $\times$ 7 days
Levofloxacin, 750 qd 5-7 days	Amoxicillin, 500 mg PO bid $\times$ 7 days ^a
	Amoxicillin/clavulanic acid, 500 mg bid or 875 mg PO bid $\times$ 7 days ^a
	Cephalexin, 500 mg qid $\times$ 7 days ^a
	Cefuroxime axetil, 500 mg bid $\times$ 7 days ^a
	Ceftriaxone, 400 mg qd $\times$ 7 days ^a
Parenteral^b	
Girofloxacillin, 400 mg q12h	Morporepen, 1 g q8h
Levofloxacin, 750 mg qd	Piperacillin/tazobactam, 3.375 g q6h
Gentamicin or tobramycin, 3-5 mg/kg qd, a p.cipillin, 1 g q4h	Doripenem, 500 mg q8h
Ceftazidime, 2 g q8h	Ertapenem, 1 g qd
Ceftriaxone, 1-2 g qd	Cefepime/tazobactam, 2 g q8h
Cefotaxime, 1-2 g qd	Ceftriaxone/zivibactam, 2 g q8h

^bChange to oral therapy to complete course once condition is clinically stable.

Table 38.6 National Institutes of Health Classification of Prostatitis Syndromes

Class	Description
I	Acute bacterial prostatitis
II	Chronic bacterial prostatitis
III	Chronic pelvic pain syndrome (CPPS)
IIla	Inflammatory CPPS
	Leukocytes present in semen, in urine after prostate massage, or in expressed prostate secretions
IIlb	Noninflammatory CPPS
	Absence of leukocytes in specimens
IV	Asymptomatic inflammatory prostatitis
	Leukocytes in specimens similar to those in inflammatory CPPS, but no symptoms

Data from Krieger JN, Nyberg L Jr, Nickel JC. NIH consensus definition and classification of prostatitis. *JAMA*. 1999;282(3):236-237.

Table_38_4

Table_38_5

Table_38_6

Table 38.7 Treatment of Candiduria		
	Cystitis	Pyelonephritis
Fluconazole (if isolate is susceptible)	200-400 mg daily, 7-14 days	200-400 mg daily, 14 days
Amphotericin B deoxycholate	0.3-0.6 mg/kg daily, 1-7 days	0.5-0.7 mg/kg daily, 14 days
Amphotericin B bladder irrigation	5-50 mg/L continuous irrigation for 2-7 days	Not indicated
Flucytosine	25 mg/kg qid, 7-10 days	25 mg/kg qid, 14 days ^a

^aMay be used in combination with amphotericin B. qid, Four times a day.
Adapted from Pappas PG, Kauffman CA, Andes DR, et al. Clinical practice guideline for the management of candidiasis: 2016 update by the Infectious Diseases Society of America. *Clin Infect*

Table_38_7